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Expt 3

PROGRAM

```
clc; clear; close all;

% Specifications
Fs=2000; Rp=1; Rs=40;
Wp=[300 600]/(Fs/2);
Ws=[200 700]/(Fs/2);

% Butterworth Filter
[n1,W1]=buttord(Wp,Ws,Rp,Rs);
[b1,a1]=butter(n1,W1,'bandpass');

% Chebyshev Type-I Filter
[n2,W2]=cheb1ord(Wp,Ws,Rp,Rs);
[b2,a2]=cheby1(n2,Rp,W2,'bandpass');

% Magnitude Response
figure;
freqz(b1,a1); title('Butterworth Filter');

figure;
freqz(b2,a2); title('Chebyshev Type-I Filter');

% Group Delay
figure;
grpdelay(b1,a1); hold on;
grpdelay(b2,a2);

legend('Butterworth','Chebyshev-I');

% Pole-Zero Plot
figure;
subplot(121),zplane(b1,a1),title('Butterworth')
subplot(122),zplane(b2,a2),title('Chebyshev-I')
```

```

% Impulse Response

figure;

subplot(211),impz(b1,a1),title('Butterworth Impulse')

subplot(212),impz(b2,a2),title('Chebyshev Impulse')

% Pole Magnitude

figure;

subplot(121),stem(abs(roots(a1)),'filled')

title('Butterworth Pole Magnitude')

subplot(122),stem(abs(roots(a2)),'filled')

title('Chebyshev Pole Magnitude')

disp(['Butterworth Order = ',num2str(n1)])

disp(['Chebyshev Order = ',num2str(n2)])

```

OUTPUT

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Butterworth Order = 9
Chebyshev Order = 5

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